

A Minimal Source-Locked Local Occurrence-Edge Witness Contract

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Abstract

TPC-173 finds no qualifying local actual-occurrence edge claim in the frozen TPC-133–172 theorem corpus. This does not remove the need for an exact acceptance interface. We define a minimal source-locked local-edge witness. Every edge has the five-field archived cut address, a distinct actual-occurrence identifier, an exact nonzero rational weight, edgewise fixed $h_0 = 2$, physical-normalization lineage, and source evidence resolving to a qualifying TPC-173 claim. Every declared local cut fibre is nonempty and has exact column sum one.

A deterministic verifier proves finite internal consistency for supplied records. It keeps source-backed local support separate from actual active support and canonical/minimal representation. The committed two-edge fixture has weights $1/3, 2/3$ and is explicitly synthetic L0 only. Since TPC-173 has zero qualifying claims, no production witness can currently instantiate the contract. The production status remains not-testable; no arithmetic L2 conclusion is obtained.

1 Source contract

Let

$$\kappa(c) = (\ell(c), k(c), d_{\text{nat}}(c), j_L(c), j_K(c)) \quad (1)$$

be the exact archived address supplied by TPC-164. It identifies a frozen cut row, not a downstream occurrence [1, 2].

Definition 1.1 (Local edge record). *A production local edge record contains*

$$(\kappa(c), o, \lambda, h_0, \nu, \mathcal{S}), \quad (2)$$

where o is an actual-occurrence identifier, $\lambda \in \mathbb{Q} \setminus \{0\}$, $h_0 = 2$, ν is the physical normalization, and $\mathcal{S}$ is the source bundle. The latter contains a TPC-173 qualifying claim ID, path, canonical hash, theorem locator, formula locator and an exactly matching nonempty derivation AST.

The edge ID is a deterministic digest of the ordered pair $(\kappa(c), o)$. Thus an archive address cannot silently become an occurrence identity.

2 Local completeness

Definition 2.1 (Covered local fibre). *For a finite nonempty declared cut set U , every $c \in U$ has a finite nonempty edge family $E(c)$ satisfying*

$$\sum_{e \in E(c)} \lambda_e = 1. \quad (3)$$

Every edge source lies in U , and edge and occurrence identifiers are globally unique inside the witness.

This is the local hypothesis required by the TPC-165 gluing theorem. It is not global coverage, actual active support, or a canonical-parent theorem.

3 Finite verifier theorem

Theorem 3.1 (Internal verifier soundness). *If the TPC-174 verifier accepts a supplied witness, then:*

1. *all edge and occurrence identifiers are unique;*
2. *every edge belongs to exactly one declared five-field cut address;*
3. *every cut fibre is nonempty and satisfies (3) in exact rational arithmetic;*
4. *fixed $h_0 = 2$ and physical normalization agree edgewise; and*
5. *in production mode, every source record resolves exactly to a qualifying TPC-173 claim, including equality of the derivation AST.*

Proof. The verifier parses exact integers and reduced fractions, reconstructs the canonical edge ID, and rejects duplicate identities. It groups edges by the ordered address (1), checks nonempty coverage and computes each sum in $\mathbb{Q}$. Equality checks enforce h_0 , normalization and the full source join. Closed root, edge and source field sets reject undeclared fields, and the complete claim-boundary record is fixed false. These are precisely the asserted finite predicates. $\square$

Proposition 3.2 (No provenance or root promotion). *Verifier acceptance does not prove the mathematical truth of a cited external theorem, actual active support, canonical/minimal actual representation, or an arithmetic saving.*

Proof. The verifier checks finite records and their source joins. It contains no proof kernel for the cited theorem, no nonzero physical-coefficient test, no representation theorem and no analytic estimate. These conclusions therefore do not follow. $\square$

4 Synthetic nonvacuity and production state

The committed fixture declares one address and two distinct typed edge rows:

$$\lambda_1 = \frac{1}{3}, \quad \lambda_2 = \frac{2}{3}.$$

It passes every identity, typing and conservation check. Its evidence mode is `SYNTHETIC_L0_ONLY`; its source locators are deliberately null and its source ID is `SYNTHETIC_AXIOM_L0`. It is not a production claim.

Theorem 4.1 (Current production verdict). *For the source-locked TPC-173 inventory,*

$$\boxed{\#Q_{173} = 0, \quad \text{Status}_{\text{production witness}} = \text{NOT-TESTABLE}.} \quad (4)$$

Proof. Production mode requires each edge source to occur in the TPC-173 qualifying-claim map. That map is empty. The synthetic fixture is rejected under production mode, so it cannot change the verdict. $\square$

5 Claim firewall

The schema and verifier are finite L0 interface results. They do not supply a production local occurrence family or close either parallel H1 root. No named fixed phase, positive fixed- X L2, strict 1/400, prime-pair lower bound, or twin-prime theorem is established.

References

- [1] Liang Wang. Source-backed local-to-global gluing for formal h1 occurrence crosswalks. *TPC Research Archive*, 2026. TPC-165.
- [2] Liang Wang. Production source-claim inventory. *TPC Research Archive*, 2026. TPC-173.